

2026-06-10-SP3-P3.3-swap-design

SP3 P3.3/P3.4 — cli_agents submodule swap to caf - * forks + GitLab population (DESIGN)

Date: 2026-06-10 **Author:** READ-ONLY DESIGN subagent (no submodule swapped, no .gitmodules edit, no commit/push, no mutating gh/gitlab) **Scope:** Design + validate the SP3 P3.3 (swap 49 cli_agents/<name> submodule URLs → their vasic-digital/caf - * GitHub forks, keep upstream as a named remote) and P3.4 (populate the empty GitLab caf - * mirrors) procedures. Execution is **OPERATOR-CONFIRM** (high blast radius). **Status:** Design complete; map validated; dry-run rewrite proven. NOT executed.

1. Map validation result (Task 1)

Source: docs/caf/map.tsv (cols: name \t upstream-ssh \t github-fork-ssh \t gitlab-fork-ssh).

Check	Result
Total rows	80
Unique names	49
Duplicate rows (byte-identical)	31 — names agent-deck...fauxpilot appear 2–3×
Columns per row	4 on every row (no ragged rows)
Line endings	LF (no CRLF)
Deduped names vs .gitmodules cli_agents/* (50 entries) minus claude-code-source	IDENTICAL 1:1 (diff -q → no differences)
claude-code-source in map?	No — correctly EXCLUDED. It is in .gitmodules (own-org private repo git@gitlab.com:milos85vasic/ccode-private.git), not a third-party fork target.
cline in map?	Yes (row points git@github.com:cline/cline.git → caf-cline).
cline-bench in map?	No (see §6 — it is a <i>nested</i> submodule inside cline, not a top-level cli_agents/entry).
Map upstream col vs current .gitmodules url for all 49	

Check	Result
	0 mismatches — the map's upstream column equals the live submodule URL, confirming the swap baseline.
All 49 github-fork-ssh well-formed git@github.com:vasic-digital/ caf-<slug>.git	✓
All 49 gitlab-fork-ssh well-formed git@gitlab.com:vasic-digital/ caf-<slug>.git	✓
Fork slug == caf-<name> for every row	✓ (no slug mismatch)

FINDING — data-integrity defect (non-blocking for the swap, but should be fixed): docs/caf/map.tsv contains **31 duplicate rows** (some names triplicated). Root cause: G-1's planner (see docs/caf/Status.md) emitted its PLAN log across ~3 passes and the map was appended each pass for the early-alphabet names. The duplicates are *byte-identical*, so dedup is lossless and the **unique set is a clean 49 mapping 1:1** onto the swappable submodules. The swap procedure below dedups defensively (awk '!seen[\$0]++'), so it is correct even against the un-cleaned file, **but the map SHOULD be rewritten to 49 unique rows** before execution (separate, low-risk commit — awk '!seen[\$0]++' map.tsv > map.tsv.new).

Conclusion: The 49 unique map rows map 1:1 onto the 49 swappable cli_agents/* submodules; claude-code-source is correctly excluded; URL formedness is clean. ✓
(modulo the duplicate-rows hygiene fix).

2. Fork-content vs GitLab-empty verification (Task 2)

Sample-checked 3 names (caf-agent-deck, caf-aider, caf-crush), READ-ONLY.

GitHub forks — have content (gh repo view ... --json name,isEmpty,defaultBranch,diskUsage,parent):

Fork	isEmpty	defaultBranch	diskUsage (KiB)	parent (real fork)
vasic-digital/ caf-agent-deck	false	main	238623	asheshgoplani/ agent-deck
vasic-digital/ caf-aider	false	main	140356	Aider-AI/aider
vasic-digital/ caf-crush	false	main	26916	charmbracelet/ crush

GitLab repos — empty (glab api projects/<enc> → empty_repo):

Repo	empty_repo	default_branch
vasic-digital/caf-agent-deck	True	main
vasic-digital/caf-aider	True	main
vasic-digital/caf-crush	True	main

Conclusion (Task 2): The swap MUST point the submodule URL at the **GitHub fork** (has content, is a real fork with a parent). The **GitLab caf-*** **repo is empty** and is a **secondary push mirror to be populated** in P3.4. Confirmed. 

3. SWAP procedure (Task 3) — .gitmodules rewrite + sync + per-fork upstream remote

OPERATOR-CONFIRM at execution. Swapping 49 submodule URLs is high-blast-radius (§11.4.101: irreversible-ish in effect on every developer clone, high blast radius). Gate the first mutating run behind an explicit operator yes. Default to a plan/dry-run.

3.1 Pre-flight (read-only, §11.4.71 fetch-first)

```
cd /Volumes/T7/Projects/HelixCode
git fetch --all --prune --tags
git submodule status cli_agents/          # capture pre-state SHAs
      (evidence baseline)
awk '!seen[$0]++' docs/caf/map.tsv > /tmp/map_uniq.tsv # dedup
      defensively (49 rows)
test "$(wc -l < /tmp/map_uniq.tsv)" -eq 49 || { echo "map not 49
      unique"; exit 1; }
```

3.2 .gitmodules URL rewrite (per row)

For each of the 49 unique rows, rewrite ONLY submodule.cli_agents/<name>.url to the **GitHub fork** column. claude-code-source is never in the map, so it is never touched.

```
while IFS=$'\t' read -r name up gh gl; do
  [ -z "$name" ] && continue
  case "$name" in \#*) continue ;; esac
  git config -f .gitmodules "submodule.cli_agents/$name.url"
    "$gh"
done < /tmp/map_uniq.tsv
```

3.3 Sync the config into .git/config + each worktree, then update

```
git submodule sync --recursive          # push .gitmodules urls
      into .git/config + submodule .git/config
git submodule update --init --recursive cli_agents/ # re-point
      each working submodule at the fork
```

Because the fork is a true fork of upstream, the previously-checked-out commit SHA is present in the fork's object graph, so `git submodule update` re-points cleanly without re-clone in most cases. If a pinned SHA is NOT yet in the fork (fork behind upstream), update that fork first via `update_fork_from_upstream.sh --only <name> --execute` (P3.4 machinery) THEN re-run `git submodule update` for it.

3.4 Keep upstream as a named remote per fork (§11.4.71 merge-from-upstream)

Inside each submodule working tree, `origin` now = the `caf-*` fork. Add upstream as a second named remote so future `update_fork_from_upstream.sh` (and manual merges) can fetch upstream:

```
while IFS=$'\t' read -r name up gh gl; do
  d="cli_agents/$name"
  ( cd "$d" 2>/dev/null \
    && git remote get-url origin >/dev/null 2>&1 \
    && { git remote add upstream "$up" 2>/dev/null || git remote
      set-url upstream "$up"; } \
    && git remote add gitlab "$gl" 2>/dev/null || git remote set-
      url gitlab "$gl" )
done < /tmp/map_uniq.tsv
```

Result per submodule: `origin`→ GitHub fork (fetch+push), `upstream`→ third-party source (fetch only, for §11.4.71 merges), `gitlab`→ GitLab mirror (push). NOTE: the `upstream/gitlab` remotes live in each submodule's `.git/config` (per-clone, NOT versioned in `.gitmodules`); document them in `docs/caf/README.md` so a fresh clone re-adds them (or wire into `scripts/init-submodules.sh`).

3.5 Buildability spot-check (anti-bluff, §11.4.5 captured evidence)

The `cli_agents` submodules are *reference* agents (multi-language), not part of the Go build graph, so “buildable” = (a) submodule checks out at its pinned SHA, (b) a representative few build/parse in their own toolchain. Spot-check ≥ 3 across languages:

```
git submodule status cli_agents/ | grep -v '^-'
# no '-' prefix => all initialised (not de-init)
# Rust: ( cd cli_agents/aichat && cargo metadata --no-deps >/
dev/null ) # crush is Go; aichat is Rust
# Go: ( cd cli_agents/crush && go vet ./... 2>&1 | head )
# Python: ( cd cli_agents/aider && python -c 'import sys;
print(sys.version)' && ls pyproject.toml )
```

Capture stdout/exit codes as the evidence artefact. A swap that leaves a submodule at a SHA the fork lacks → `git submodule update` errors → that fork needs P3.4 sync first (see §3.3).

3.6 Rollback (reversibility — §11.4.101)

The whole swap is one `.gitmodules` edit + config sync. To revert before commit: `git checkout .gitmodules && git submodule sync --recursive && git`

```
submodule update --init --recursive cli_agents/. Keep a copy of pre-swap
git submodule status cli_agents/ as the known-good baseline.
```

4. GitLab-population step (Task 4) — P3.4

The 49 `caf - *` GitLab repos are `empty_repo=True`. Populate each by mirroring the **fork content** (the GitHub `caf - *` fork already holds full upstream history) into the GitLab repo — **fast-forward only, NEVER force** (§11.4.113).

Reuse the existing machinery (CONST-074/§11.4.74 — extend, don't reimplement):

`scripts/caf/update_fork_from_upstream.sh` already: clones the GitHub fork (origin), adds upstream, fetches, merges upstream `main/master` (ff-preferred, conflicts parked, never -s ours), then pushes to **all provider remotes** via `caf_provider_remotes` → `caf_safe_push`. `caf_safe_push/caf_assert_no_force refuse --force/--force-with-lease/--mirror/+ref` (§11.4.113-safe by construction). Providers come from `CAF_PROVIDERS` csv (default includes `github, gitlab`), so a single run also seeds GitLab.

Because the GitLab side is *empty*, the first push to `gitlab` is a fresh-branch push (still a fast-forward against an empty ref — no history to overwrite, no force needed):

```
# DRY-RUN (default, mutate nothing) — plan the merges + pushes:
bash scripts/caf/update_fork_from_upstream.sh --map-file /tmp/
    map_uniq.tsv
```

```
# OPERATOR-CONFIRM execute (G-1-push gate, §11.4.101): seeds
    GitHub fork (idempotent ff)
# AND populates the empty GitLab mirror, fast-forward only:
bash scripts/caf/update_fork_from_upstream.sh --map-file /tmp/
    map_uniq.tsv --execute
```

Notes: - The script's `caf_detect_default_branch` resolves `main/master` from the upstream symref per fork (handles mixed default branches). - Conflicts are parked to `docs/caf/conflicts/<fork>_<ts>.md` and the loop continues (zero-idle); none expected for an initial seed. - For forks whose pinned submodule SHA was missing (see §3.3), this run brings the fork current first, unblocking §3.3 `git submodule update`. - If a leaner “seed only” is preferred over merge-from-upstream, an equivalent is a per-fork `git push gitlab --all && git push gitlab --tags` from a fork clone — still ff-only, no `--mirror` (mirror is force-class and is blocked). The existing script is preferred (already anti-bluff + §11.4.113-guarded + has a Challenge).

5. Dry-run .gitmodules rewrite diff sample (Task 5)

Performed against a **TEMP copy** (`/tmp/gitmodules.swap`), live `.gitmodules` untouched, then the temp was deleted.

```
$ cp .gitmodules /tmp/gitmodules.swap
$ while IFS=$'\t' read -r name up gh gl; do
    git config -f /tmp/gitmodules.swap "submodule.cli_agents/$name.url" "$
done < <(awk -F'\t' '!seen[$0]++' docs/caf/map.tsv)
```

Results: - Changed url = lines: **49** (exactly the 49 swappable submodules). - claude-code-source url AFTER rewrite: git@gitlab.com:milos85vasic/ccode-private.git — **unchanged** (correctly excluded). - Symmetry: 49 removed (<) [?] 49 added (>); **every** added line is a caf-* fork (grep -v 'caf-' on > lines → none). - Safety: no changed line outside cli_agents/ (no Go-app / HelixAgent / qa submodule touched).

Diff sample (unified, first rows):

```
--- .gitmodules
+++ /tmp/gitmodules.swap
@@ -63,154 +63,154 @@
 [submodule "cli_agents/agent-deck"]
     path = cli_agents/agent-deck
-    url = git@github.com:asheshgopani/agent-deck.git
+    url = git@github.com:vasic-digital/caf-agent-deck.git
 [submodule "cli_agents/aiagent"]
     path = cli_agents/aiagent
-    url = git@github.com:Xiaoccer/AIAgent.git
+    url = git@github.com:vasic-digital/caf-aiagent.git
 [submodule "cli_agents/aichat"]
     path = cli_agents/aichat
-    url = git@github.com:sigoden/aichat.git
+    url = git@github.com:vasic-digital/caf-aichat.git
 [submodule "cli_agents/aider"]
     path = cli_agents/aider
-    url = git@github.com:Aider-AI/aider.git
+    url = git@github.com:vasic-digital/caf-aider.git
 [submodule "cli_agents/amazon-q"]
     path = cli_agents/amazon-q
-    url = git@github.com:aws/amazon-q-developer-cli.git
+    url = git@github.com:vasic-digital/caf-amazon-q.git
```

Temp file removed after capture. The rewrite is **correct**: 49 url lines, scoped to cli_agents/, fork URLs only, claude-code-source preserved.

6. Nested cline → cline-bench (handled)

cli_agents/cline/.gitmodules declares a **nested** submodule:

```
[submodule "evals/cline-bench"]
    path = evals/cline-bench
    url = https://github.com/cline/cline-bench.git
```

Two facts: 1. It is **NOT** a top-level cli_agents/ entry, so it is **correctly absent** from map.tsv (the map only forks the 49 top-level reference agents). 2. Its URL is **HTTPS** — a **Rule 3 violation** if/when initialized, and a CONST-051(C) **nested-own-org-chain** concern *only if* it pointed at an own-org repo (it points at third-party cline/cline-bench, so the nested-chain ban does not apply — third-party nested submodules are exempt per CONST-051(C)).

Recommendation for P3.3/P3.4 (own scope, not blocking the 49-swap): - Do **not** recursively init evals/cline-bench during the swap (`git submodule update` for `cli_agents/cline` should NOT pass `--recursive` into cline's own nested tree, or should skip it) to avoid pulling an HTTPS submodule into the tree. - If `cline-bench` is needed, fork it to `vasic-digital/caf-cline-bench` and add it to the map as a **50th row** + place it at the project root per CONST-051(C) (flat `<root>/caf-cline-bench` or grouped), then have cline reference it via config injection — but this is a **separate, operator-gated work item**, NOT part of the 49-swap. Track it as its own ticket.

7. Summary

Map validation: `docs/caf/map.tsv` has 80 rows but **49 unique** (31 byte-identical duplicates — a hygiene defect to clean to 49 rows). The 49 unique names map **1:1** with the 49 swappable `cli_agents/*` submodules; `claude-code-source` excluded; `cline` present, `cline-bench` correctly absent (nested-in-cline, not top-level); all 49 fork/mirror URLs well-formed; `map upstream col == live .gitmodules url` (0 mismatch). 

Swap procedure: `fetch-first` → dedup map to 49 → `git config -f .gitmodules submodule.cli_agents/<name>.url <github-fork>` per row → `git submodule sync --recursive` → `git submodule update --init --recursive cli_agents/` → add upstream+gitlab named remotes per submodule (for §11.4.71 merges; per-clone, document in README) → buildability spot-check (Rust/Go/Python) with captured evidence → `rollback = git checkout .gitmodules + re-sync`.
OPERATOR-CONFIRM at execution (49 URLs = high blast radius).

GitLab population: reuse `scripts/caf/update_fork_from_upstream.sh --map-file <49-row map> --execute` — it merges upstream into the fork (ff-preferred, conflicts parked, never -s ours) and pushes to all providers (`github`, `gitlab`) via `caf_safe_push` which **refuses force-class args (§11.4.113)**; the empty GitLab mirror seeds as a fast-forward fresh-branch push (no force, no `--mirror`). Default is dry-run; the first mutating push is operator-gated.

Dry-run diff (proof): TEMP-copy rewrite changed **exactly 49** url lines, all `cli_agents/*`, all → `caf-*` forks, `claude-code-source` untouched, 49[?]49 symmetric, nothing outside `cli_agents/` changed. Temp deleted. The rewrite is correct.

Open items (separate tickets): (a) clean `map.tsv` to 49 unique rows; (b) decide `cline-bench` fork/exclude policy (HTTPS + nested).